

Regularization of Restricted 3-Body Problems

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Abstract. Using geometric mechanics we regularize the binary collisions of planar restricted three-body problems (with zero and nonzero angular momentum). The regularization is general enough to allow the primaries to move under any arbitrary dynamics and has as special cases all the classical restricted three-body problems, namely; circular, elliptic, parabolic and hyperbolic. This generalizes the classical procedures of Levi-Civita, Birkhoff, Thiele and Burrau. Explicit formulas are given for the regularizations.

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1. Introduction

Regularization of collisions in celestial mechanics systems is one of the main tools to study close approaches between the bodies. Levi-Civita noticed that for the planar two-body problem (Kepler's problem) the singularities of motion due to collisions between the bodies could be removed by doing proper coordinate and parameter changes. The procedure was generalized to the three-dimensional problem by the so called Kustanheimo–Stiefel transformation (see [26]). Regularization was soon applied to the circular restricted three-body problem. Since in the circular restricted three-body problem we have two possible binary collisions there are essentially two ways of approaching the problem. The first is to regularize only one of the possible collisions (the local approach) and the second is to regularize both collisions at the same time (the global approach). For extensive descriptions and references on the regularization problem see [28] and [26]. It is important to note here that the regularization is deeply related with the problem of classification of the final evolutions of the three-body problems as noted by Chazy in [9].

In this work we study the regularization of a general class of restricted three-body problems (with zero and nonzero angular momentum). We offer a geometric

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description of the regularization procedure that is both simple and general. Our formalism is broad enough to include any dynamics for the primaries relative motion (e.g drag effects). Assuming only that the primaries do not collide among themselves and that they attract the infinitesimal body according Newton's law we regularize locally and globally the binary collisions. Our main interest is to regularize the parabolic and hyperbolic restricted three-body problems [11–13]. Since for these restricted three-body problems the distance between the primaries goes to infinity with time one could ask why to regularize them? We answer this question discussing the problem that motivated our regularization, namely the problem of capture in the parabolic and hyperbolic restricted three-body problems: Assume that in the parabolic (or hyperbolic) restricted three-body problem one of the primaries comes from $t = -\infty$ carrying the zero mass body ($m_3 = 0$). Assume also that after the approach between the primaries the zero mass body is captured by the other primary in such a way to follow it for $t = +\infty$ (i.e. the zero mass body goes from one primary to the other). We call an orbit of this type a capture orbit [2, 4, 8–10, 15, 16, 20–24]. We would like to characterize the set of all capture orbits. In [4] Alvarez, Delgado and Cors showed that the Parabolic restricted three-body problem is a gradient-like dynamical system. As a corollary it follows that a large subset of the solutions (in a measure theoretic sense) tends to a collisional singularity when $t \rightarrow -\infty$ and when $t \rightarrow +\infty$. In other terms, their theorem shows that capture orbits are close to ejection-collision solutions. The main idea follows from the fact that the regularizations made in this work map the collisions in well defined points of the regularized configuration space. Thus understanding the set of ejection-collision solutions (therefore the set of capture orbits) will be equivalent to solve a two-point boundary value problem in the regularized system.

According [3] there are exactly 30 different restricted three-body problems in dimension 1, 2 and 3. At least 10 of these problems have been studied by several authors. Moreover, we observe that for the restricted three-body problems of dimension one (namely, parabolic, hyperbolic and elliptic) the study of the global flow is not simple, mainly because those problems are not conservative.

Classical regularization in Celestial Mechanics uses essentially the fact that the energy is conserved through the motion. Regularization consist mainly in considering a suitable change of coordinates and a time transformation. The time transformation depends on the conservative character of the motion (see Theorem 3 above). It follows that the classical regularization approach results inappropriate when one considers a non conservative system. Here is located our main contribution. In this work the main role is played by the Hamiltonian character of the system in the *extended* phase space. The geometric formulation (see e.g. [7]) of the regularization procedure allow us to regularize all planar restricted three-body problems in a unique and global way. The regularization is general enough to regularize all restricted three-body problems in only one procedure. The global regularization is important for the study of the ejection-collision solutions since it fixes the collision points in the regularized configuration space. Numerically it reduces the study of ejection-collision solutions to a two-point boundary value

problem and opens the possibility of investigating the capture problem in a systematic and efficient way.

We outline the article. In Section 2 we review the Levi-Civita regularization of the two-body problem using the language of geometric mechanics. In section 3 we regularize locally general restricted three-body problems using the tolls introduced in Section 2. In Section 4 we regularize globally the general restricted three-body problem using a generalization of the Birkhoff and the Thiele–Burrau transformations. Formulas are explicitly computed in order to facilitate numerical implementations. In Section 5, as an illustration of our method we regularize the one-dimensional parabolic restricted three-body problem, we compute the local regularization and the Birkhoff regularization (global regularization).

2. Hamiltonian regularization of the Kepler problem

We review the Levi-Civita regularization under the eyes of geometric mechanics (for a general treatment of geometric mechanics as well as the demonstrations of theorems (1) and (3) see e.g. [1, 19]). The theorems introduced in this section will be stated without proofs. Let M be a symplectic manifold and w its symplectic two-form. In celestial mechanics problems M is usually the cotangent bundle of a configuration space C (in this paper C will be $\mathbf{R}^2$ minus the set of the singularities, then C is an open subset of $\mathbf{R}^2$), i.e., $M = T^*C$ (in this paper $M = C \times \mathbf{R}^2$) and w is the standard symplectic two-form. The following theorem states that any transformation of coordinates in C generates a canonical transformation (symplectomorphism) of T^*C . It is the bundle version of the well known result that contact transformations induces canonical transformations in phase space.

Theorem 1. *Let $\phi : C \rightarrow C$ be a diffeomorphism. Then $\phi : T^*C \rightarrow T^*C$, $\phi = (d\phi^T)^{-1}$ is a canonical transformation. Here $d\phi^T$ is the adjoint of the derivative of ϕ .*

Example (The Levi-Civita transformation). For the planar Kepler problem the configuration space is given by $C = \mathbf{R}^2 - \{(0, 0)\}$ and the phase space is given by T^*C . Since we will be considering time reparametrizations it is convenient to work in the extended phase space. This is nothing more than the usual phase space direct product with $\mathbf{R}^2$. Physically this means that energy E and time t are included as canonically conjugated variables, i.e., our configuration space will be given by $\tilde{C} = C \times \mathbf{R}$ and parametrized by (q_1, q_2, t) . The phase space will be given by $T^*\tilde{C}$ and parametrized by $(q_1, q_2, t, p_1, p_2, E)$. The symplectic two form is the canonical one: $w = dq_1 \wedge dp_1 + dq_2 \wedge dp_2 + dt \wedge dE$. The Levi-Civita transformation $\phi : C \rightarrow C$ is given by the parabolic transformation $(q_1, q_2) = \phi(u, v) = (u^2 - v^2, 2uv)$. Looking C as imbedded in $\tilde{C}$ this induces the transformation $\tilde{\phi}(u, v, t) = (u^2 - v^2, 2uv, t)$. Therefore

$$d\tilde{\phi} = \begin{pmatrix} 2u & -2v & 0 \\ 2v & 2u & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (1)$$

and

$$(d\tilde{\phi}^T)^{-1} = \frac{1}{4(u^2 + v^2)} \begin{pmatrix} 2u & -2v & 0 \\ 2v & 2u & 0 \\ 0 & 0 & 4(u^2 + v^2) \end{pmatrix}. \quad (2)$$

The canonical transformation $\tilde{\phi} = \tilde{\phi}(u, v, t, p_u, p_v, E)$ is given by

$$\tilde{\phi} = \left(u^2 - v^2, 2uv, t, \frac{up_u - vp_v}{2|\xi|^2}, \frac{vp_u + up_v}{2|\xi|^2}, E \right)$$

where $|\xi|^2 = u^2 + v^2$. The Hamiltonian function of the Kepler problem is given by

$$H = \frac{p_1^2}{2} + \frac{p_2^2}{2} - \frac{\kappa}{r}$$

where $r = \sqrt{q_1^2 + q_2^2}$ and κ is a positive constant. Observe that under the transformation $\tilde{\phi}$ we have

$$\tilde{H}(u, v, t, p_u, p_v, E) = H(\tilde{\phi}(q_1, q_2, t, p_1, p_2, E)) = \frac{p_u^2 + p_v^2}{8|\xi|^2} - \frac{\kappa}{|\xi|^2}.$$

The important fact here is that the Hamiltonian function becomes homogeneous of degree -2 in $|\xi|$. We will call $\xi = (u, v)$ and $p_\xi = (p_u, p_v)$.

Remark 2. We use the term $dt \wedge dE$ instead of the common term $dE \wedge dt$. The two different choices correspond to different parametrizations of the Hamiltonian solutions (doing $t \rightarrow -t$ we change the form $dt \wedge dE$ into $dE \wedge dt$). This will become clear in the next example.

The following theorem relates the reparametrization procedure with the hypersurface of constant energy defined by the Hamiltonian function.

Theorem 3. *Let H, F be two Hamiltonians such that $\{H = E_h\} = \{F = E_f\}$ as sets, where $E_h, E_f \in \mathbf{R}$. Then, the Hamiltonian flow generated by H at the level E_h is equal to the Hamiltonian flow of F at the level E_f up to reparametrization.*

Example (The regularized Kepler problem). Consider the Hamiltonians $\tilde{H}$ and $\mathcal{H} = |\xi|^2 (\tilde{H} - E)$. We have that as sets $\{\tilde{H} = E\} = \{\mathcal{H} = 0\}$. Observe also, that $\xi = 0$ is a singularity for $\mathcal{H}$. By the theorem, the induced flows at the respective levels are equal up to reparametrization. In fact writing Hamilton's equations for $\mathcal{H}$ we obtain that

$$\begin{aligned} \frac{d\xi}{ds} &= |\xi|^2 \frac{\partial \tilde{H}}{\partial p_\xi}, \\ \frac{dp_\xi}{ds} &= -2\xi (\tilde{H} - E) - |\xi|^2 \frac{\partial \tilde{H}}{\partial \xi}, \\ \frac{dt}{ds} &= -|\xi|^2, \\ \frac{dE}{ds} &= 0. \end{aligned} \quad (3)$$

At $\mathcal{H} = 0$, i.e., at $\bar{H} = E$ it follows that the equations read as

$$\begin{aligned}\frac{d\xi}{ds} &= |\xi|^2 \frac{\partial \bar{H}}{\partial p_\xi}, \\ \frac{dp_\xi}{ds} &= -|\xi|^2 \frac{\partial \bar{H}}{\partial \xi}, \\ \frac{dt}{ds} &= -|\xi|^2, \\ \frac{dE}{ds} &= 0.\end{aligned}\tag{4}$$

The third equation is just a time reparametrization (the Sundman reparametrization). Replacing $t \rightarrow -t$ (see the remark above) we rewrite those equations as

$$\begin{aligned}\frac{d\xi}{dt} &= \frac{\partial \bar{H}}{\partial p_\xi}, \\ \frac{dp_\xi}{dt} &= -\frac{\partial \bar{H}}{\partial \xi};\end{aligned}\tag{5}$$

but those are Hamilton's equations for H at the level set E . Observe that equations (4) are not singular at $\xi = 0$.

Remark 4. The parabolic transformation was introduced by Levi-Civita (see [18] and references therein). Under this transformation the motion is prolonged through collision as an elastic bouncing. The time reparametrization was introduced by Sundman [27]. The new parameter is interpreted as the eccentric anomaly.

3. Local regularization

Local regularization can be achieved translating the origin of the coordinate system to one of the primaries and then applying a Levi-Civita procedure as made in the previous section. Let $\vec{r}(t)$ be the relative vector between the two primaries of mass $\mu_1 = \mu$ and $\mu_2 = 1 - \mu$ where $0 < \mu < 1$. The only restriction we will impose on $\vec{r}(t)$ is that $|\vec{r}(t)| \neq 0$. For Newtonian restricted three-body problems $\vec{r}(t)$ is a solution of Kepler's equation. Call $\vec{q}$ the position vector of the zero mass body with respect to the center of the mass of the system. We define

$$\begin{aligned}\vec{r}_1 &= \vec{q} - (1 - \mu) \vec{r}, \\ \vec{r}_2 &= \vec{q} + \mu \vec{r}.\end{aligned}$$

Let $\vec{r}(t) = (r(t), s(t))$ and $\vec{q} = (x, y)$ (see Figure 1). The dynamic of the zero mass body is given by the time dependent Hamiltonian system defined in the extended phase space with Hamiltonian function H and symplectic 2-form w given

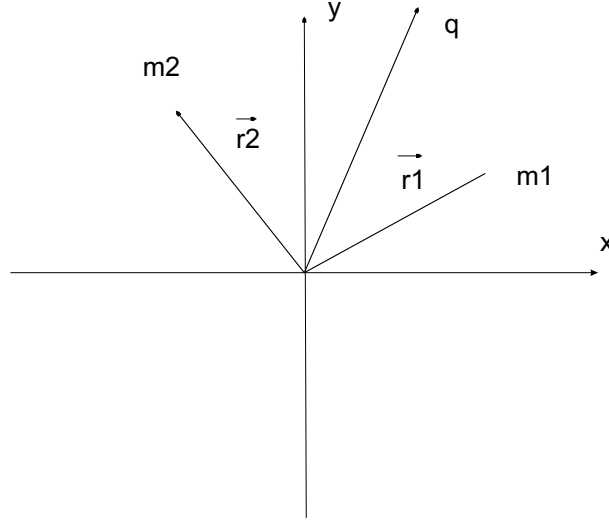


FIGURE 1. Restricted 3-body problem.

by

$$\begin{aligned}
 H &= \frac{p_x^2}{2} + \frac{p_y^2}{2} - \frac{\mu_1}{\sqrt{(x - \mu_2 r(t))^2 + (y - \mu_2 s(t))^2}} \\
 &\quad - \frac{\mu_2}{\sqrt{(x + \mu_1 r(t))^2 + (y + \mu_1 s(t))^2}} - E_t, \\
 w &= dx \wedge dp_x + dy \wedge dp_y + dt \wedge dE_t.
 \end{aligned} \tag{6}$$

The independent parameter is denoted by s . In order to regularize the binary collision between the infinitesimal mass and the body with mass μ_2 we do the contact transformation

$$\begin{aligned}
 x &= u^2 - v^2 - (1 - \mu) r(\tau), \\
 y &= 2uv - (1 - \mu) s(\tau), \\
 t &= \tau.
 \end{aligned}$$

This transformation extends to the full phase space giving

$$\begin{aligned}
 p_x &= \frac{(u p_u - v p_v)}{2(u^2 + v^2)}, \\
 p_y &= \frac{(v p_u + u p_v)}{2(u^2 + v^2)}, \\
 E_t &= \frac{(1 - \mu)}{2(u^2 + v^2)} \left\{ (v p_u + u p_v) \frac{ds}{d\tau} + (u p_u - v p_v) \frac{dr}{d\tau} \right\} + E_\tau.
 \end{aligned}$$

Under this transformation the Hamiltonian system (6) becomes

$$H = \frac{1}{4(u^2 + v^2)} \mathcal{H} \quad (7)$$

with

$$\mathcal{H} = \frac{p_u^2}{2} + \frac{p_v^2}{2} - 4\mu - \frac{4(1-\mu)(u^2 + v^2)}{\sqrt{(u^2 - v^2 - r)^2 + (2uv - s)^2}} - F, \quad (8)$$

where

$$\begin{aligned} & F(u, p_u, v, p_v, \tau, E_\tau) \\ &= 2(1-\mu) \left\{ (vp_u + up_v) \frac{ds}{d\tau} + (up_u - vp_v) \frac{dr}{d\tau} + \frac{2(u^2 + v^2)}{(1-\mu)} E_\tau \right\}, \end{aligned}$$

with symplectic 2-form given by

$$w = du \wedge dp_u + dv \wedge dp_v + d\tau \wedge dE_\tau. \quad (9)$$

Equation (7) shows that the Hamiltonian vector field induced by H is a time reparametrization of the one induced by $\mathcal{H}$. But $\mathcal{H}$ is regular at the collision, i.e., when $u, v \rightarrow 0$. Therefore the regularized system is given by Hamiltonian (8) with symplectic 2-form (9). The time reparametrization is given by

$$\frac{d\eta}{ds} = 4(u^2 + v^2),$$

where η is the new independent parameter.

4. Global regularization

To regularize both collisional singularities of system (6) simultaneously we introduce a moving system of coordinates. The x -axis is defined as the oriented line containing the two primaries. This can be done using a time dependent rotation. After the rotation is performed (and its lift to the cotangent bundle computed) we scale the system by a time dependent factor so that the distance between the primaries is constant and equal to 1.

First we define $\theta = \theta(t)$ as the angle such that

$$\cos(\theta) = \frac{r(t)}{\sqrt{s(t)^2 + r(t)^2}},$$

and

$$\sin(\theta) = \frac{s(t)}{\sqrt{s(t)^2 + r(t)^2}}.$$

We do the time dependent rotation given by

$$\begin{aligned}x &= \cos(\theta) \tilde{x} - \sin(\theta) \tilde{y}, \\y &= \sin(\theta) \tilde{x} + \cos(\theta) \tilde{y}, \\t &= \tilde{t}.\end{aligned}$$

This transformation lifts to the cotangent bundle as

$$\begin{aligned}p_x &= \cos(\theta) p_{\tilde{x}} + \sin(\theta) p_{\tilde{y}}, \\p_y &= -\sin(\theta) p_{\tilde{x}} + \cos(\theta) p_{\tilde{y}}, \\E_t &= -\tilde{y} \dot{\theta} p_{\tilde{x}} + \tilde{x} \dot{\theta} p_{\tilde{y}} + \tilde{E}_{\tilde{t}};\end{aligned}$$

where $\cdot = \frac{\partial}{\partial t}$. Under this transformation Hamiltonian system (6) becomes

$$\begin{aligned}H &= \frac{p_{\tilde{x}}^2 + p_{\tilde{y}}^2}{2} - \frac{\mu_1}{\sqrt{(\tilde{x} - \mu_2 R(\tilde{t}))^2 + \tilde{y}^2}} - \frac{\mu_2}{\sqrt{(\tilde{x} - \mu_1 R(\tilde{t}))^2 + \tilde{y}^2}} \\&\quad + (\tilde{y} p_{\tilde{x}} - \tilde{x} p_{\tilde{y}}) \dot{\theta} + \tilde{E}_{\tilde{t}}, \\w &= d\tilde{x} \wedge dp_{\tilde{x}} + d\tilde{y} \wedge dp_{\tilde{y}} + d\tilde{t} \wedge d\tilde{E}_{\tilde{t}};\end{aligned}$$

where $R = \sqrt{r^2 + s^2}$ is the distance between the primaries.

Remark 5. Observe that if $\dot{\theta} = c$, a constant, and $R(\tilde{t}) = 1$ this system reduces to the circular restricted three-body problem.

We now do the time dependent scaling

$$\begin{aligned}\tilde{x} &= R(\tilde{t}) \bar{x}, \\\tilde{y} &= R(\tilde{t}) \bar{y}, \\\tilde{t} &= \bar{t}.\end{aligned}$$

This lifts to the cotangent bundle as

$$\begin{aligned}p_{\tilde{x}} &= \frac{p_{\bar{x}}}{R(\tilde{t})}, \\p_{\tilde{y}} &= \frac{p_{\bar{y}}}{R(\tilde{t})}, \\E_{\tilde{t}} &= -(\bar{x} p_{\bar{x}} + \bar{y} p_{\bar{y}}) \frac{\dot{R}}{R^2} + E_{\bar{t}}.\end{aligned}$$

Under this transformation, a time reparametrization given by

$$\frac{d\bar{s}}{ds} = \frac{1}{R^2}$$

and a translation given by (we used the same symbol to simplify notation)

$$\tilde{x} \rightarrow \tilde{x} + \left(\frac{1}{2} - \mu\right)$$

the Hamiltonian system write as

$$H = \frac{p_{\bar{x}}^2 + p_{\bar{y}}^2}{2} - \frac{R\mu}{\sqrt{(\bar{x} - \frac{1}{2})^2 + \bar{y}^2}} - \frac{R(1-\mu)}{\sqrt{(\bar{x} + \frac{1}{2})^2 + \bar{y}^2}} - R E_{\bar{t}} + \mathcal{F}, \quad (10)$$

$$w = d\bar{x} \wedge dp_{\bar{x}} + d\bar{y} \wedge dp_{\bar{y}} + d\bar{t} \wedge dE_{\bar{t}},$$

where

$$\mathcal{F} = \left(\bar{y} R \dot{\theta} + \left(\bar{x} + \frac{1}{2} - \mu \right) \frac{\dot{R}}{R} \right) p_{\bar{x}} - \left(\left(\bar{x} + \frac{1}{2} - \mu \right) R \dot{\theta} - \bar{y} \frac{\dot{R}}{R} \right) p_{\bar{y}}.$$

The independent parameter is denoted by $\bar{s}$. We remark again that for the circular restricted three-body problem we would have $R = 1$ and $\dot{\theta} = 1$. For this case (10) reduces to the usual Hamiltonian for the restricted circular three-body problem. We also observe that if $R(\bar{t}) \neq 0$ for $\bar{t} \geq 0$ (i.e., if the motion of the primaries does not go through a collision) and if $\dot{R}$ and $\dot{\theta}$ are finite for all times the Hamiltonian function (10) has no other singularities than the collisional ones. Therefore all classical procedures that regularize the circular restricted three-body problem will in fact regularize our system.

We compute the regularized system for two of the classical regularizations: the Birkhoff and the Thiele–Burrau which were introduced in ([14, 29] and [6]) and [5] respectively. Cotangent lifts and time reparametrizations are given.

4.1. Birkhoff regularization

Birkhoff regularization is achieved doing the complex transformation

$$q = \frac{1}{4} \left(2w + \frac{1}{2w} \right).$$

We recall that the independent parameter is denoted by $\bar{s}$. writing $q = \tilde{x} + i\tilde{y}$ and $w = u + iv$ we do the transformation in extended configuration space given by

$$\begin{aligned} \bar{x} &= \frac{u}{2(u^2 + v^2)} \left(u^2 + v^2 + \frac{1}{4} \right), \\ \bar{y} &= \frac{v}{2(u^2 + v^2)} \left(u^2 + v^2 - \frac{1}{4} \right), \\ \bar{t} &= \tau. \end{aligned} \quad (11)$$

The cotangent lift is given by

$$\begin{aligned} p_{\bar{x}} &= \frac{8}{D} [4(u^2 + v^2)^2 + v^2 - u^2] p_u - \frac{16}{D} uv p_v, \\ p_{\bar{y}} &= \frac{16}{D} uv p_u + \frac{8}{D} [4(u^2 + v^2)^2 + v^2 - u^2] p_v, \\ E_{\bar{t}} &= E_{\tau}; \end{aligned}$$

where

$$D = \frac{(4(u^2 + v^2)^2 + v^2 - u^2)^2 + 4u^2 v^2}{(u^2 + v^2)^2} = 16(u^2 + v^2)^2 + 8v^2 - 8u^2 + 1.$$

Doing the time parametrization

$$\frac{d\eta}{d\bar{s}} = \frac{1}{16} \left[\frac{(1 + 4(u^2 + v^2) - 4u)(1 + 4(u^2 + v^2) + 4u)}{4(u^2 + v^2)^2} \right]$$

we obtain after some algebra that the regularized Hamiltonian system is given by

$$\begin{aligned} H &= \frac{p_u^2}{2} + \frac{p_v^2}{2} - \frac{R}{8(u^2 + v^2)^{\frac{3}{2}}} \left\{ (\mu - 1)[(1 - 2\mu)^2 + 4v^2] \right. \\ &\quad \left. - [(1 + 2u^2) + 4v^2] \right\} + \mathcal{J}, \\ w &= du \wedge dp_u + dv \wedge dp_v + d\tau \wedge dE_\tau \end{aligned} \quad (12)$$

where

$$\mathcal{J} = \frac{d\eta}{d\bar{s}} (-R^2 E + R\mathcal{F}).$$

The new parameter is denoted by η .

Remark 6. We observe that Birkhoff regularization removes the singularities due to collisions but introduces a new singularity at the origin corresponding to the physical infinity. The correspondence between the points of the q plane and of the w plane is one-to-two, in fact, $4w^2 - 8wq + 1 = 0$, and therefore $w_1 w_2 = \frac{1}{4}$. This means that two values of w corresponding to a single value of q are separated by the real axis of the w plane and, if $|w_1| \leq \frac{1}{2}$, thus $|w_2| \geq \frac{1}{2}$ and vice-versa. While the geometric mean of the two values of w corresponding to the same q is $\frac{1}{2}$, $(w_1 w_2)^{\frac{1}{2}} = \frac{1}{2}$, their arithmetic mean is q since $\frac{w_1 + w_2}{2} = q$. See [28] for more details.

4.2. Thiele–Burrau regularization

Thiele–Burrau regularization is achieved by the complex transformation given by

$$q = \frac{1}{2} \cos(w).$$

Writing $q = x + iy$ and $w = u + iv$ we write

$$\begin{aligned} \bar{x} &= \frac{1}{2} \cos(u) \cosh(v), \\ \bar{y} &= -\frac{1}{2} \sin(u) \sinh(v), \\ \bar{t} &= \tau. \end{aligned}$$

The cotangent lift is given by

$$\begin{aligned} p_{\bar{x}} &= \frac{\sin u \cosh v}{D} p_u - \frac{\cos u \sinh v}{D} p_v, \\ p_{\bar{y}} &= \frac{\cos u \sinh v}{D} p_u + \frac{\sin u \cosh v}{D} p_v, \\ E_{\bar{t}} &= E_{\tau}; \end{aligned}$$

where

$$D = \frac{(\cos u)^2 - (\cosh v)^2}{2}.$$

Doing the time parametrization given by

$$\frac{d\eta}{d\bar{s}} = \frac{1}{8} [\cosh(2v) - \cos(2u)]$$

we obtain that the regularized Hamiltonian system is given by

$$\begin{aligned} H &= \frac{p_u^2}{2} + \frac{p_v^2}{2} - \frac{R}{8} \left\{ -\mu (\cosh(v) + \cos(u)) - 2(1-\mu)(\cosh(v) \right. \\ &\quad \left. - \cos(u)) \right\} + (R^2 E + R\mathcal{J}) \frac{1}{8} (\cosh(2v) - \cos(2u)), \\ w &= du \wedge dp_u + dv \wedge dp_v + d\tau \wedge dE \end{aligned} \quad (13)$$

where

$$\mathcal{J} = \frac{d\eta}{d\bar{s}} (-R^2 E + R\mathcal{F}).$$

The new independent parameter is denoted by η .

Remark 7. The relation $q = \frac{e^{iw} + e^{-iw}}{4}$ exhibits a close relation with $q = \frac{1}{4}(2w + \frac{1}{2w})$. In fact if $w = w_B$ in the last equation where the subscript B refers to the Birkhoff's transformation, and if $w = w_T$ is the first equation, then we have $w_B = \frac{1}{2}e^{iw_T}$.

5. Example

In order to exemplify our previous approach we apply the local regularization and global regularization (Birkhoff's regularization) to a very simple example: the one dimensional Newtonian parabolic restricted three-body problem. This is a fascinating problem. In fact, one can show by appropriate change of coordinates and parameter that the system is isomorphic to an autonomous and gradient-like dynamical system. We assume the primaries are being ejected from a collision at time $t = 0$ and have total energy $h = 0$. We also assume that after the ejection the mass $m_1 = (1 - \mu)$ is at the right hand side of the binary. We consider an infinitesimal mass $m_3 = 0$ at the right of both primaries at time $t = 0$ (see Figure 2).

Our goal is to write the regularized equations of motion in both cases. We will assume units such that $G = 1$. Let $r(t)$ denote the distance between the primaries.



FIGURE 2. Parabolic restricted three-body.

The Hamiltonian for this restricted problem, written with respect to the center of mass of the primaries, is given by

$$H = \frac{p_x^2}{2} - \frac{\mu}{x - (1 - \mu)r(t)} - \frac{(1 - \mu)}{x + \mu r(t)} - E. \quad (14)$$

5.1. Local regularization

Following the notation used in Subsection 3 we must apply a Levi-Civita procedure. In order to regularize the binary collision between the infinitesimal mass and the body with mass μ_1 in this case (one dimensional) the new variables are given by

$$\begin{aligned} x &= u^2 - (1 - \mu)r(\tau), \\ t &= \tau, \\ p_x &= \frac{p_u}{2u}, \\ E_t &= \frac{(1 - \mu)}{2u} p_u \frac{dr}{d\tau} + E_\tau. \end{aligned}$$

Under this transformation the new Hamiltonian system associated to (14) becomes

$$\mathcal{H} = \frac{p_u^2}{2} - 4\mu - \frac{4(1 - \mu)u^2}{(u^2 - r)} - F, \quad (15)$$

where

$$F(u, p_u, \tau, E_\tau) = 2(1 - \mu)u \left\{ p_u \frac{dr}{d\tau} + \frac{2u}{(1 - \mu)} E_\tau \right\}.$$

5.2. Global regularization

Using Birkhoff regularization (as in Subsection 4.1) we introduce the transformation

$$\begin{aligned} x &= r(\bar{t})\bar{x}, \\ t &= \bar{t}, \end{aligned}$$

with lift

$$\begin{aligned} p_x &= \frac{1}{r(\bar{t})} p_{\bar{x}}, \\ E &= -\bar{x} \frac{\dot{r}(\bar{t})}{r(\bar{t})} p_{\bar{x}} + E_{\bar{t}}. \end{aligned}$$

Hamiltonian (14) transforms to

$$H = \frac{1}{r(\bar{t})^2} \frac{p_{\bar{x}}^2}{2} - \frac{1}{r(\bar{t})} \frac{\mu}{(\bar{x} - (1 - \mu))} - \frac{1}{r(\bar{t})} \frac{(1 - \mu)}{(\bar{x} + \mu)} + \bar{x} \frac{\dot{r}(\bar{t})}{r(\bar{t})} p_{\bar{x}} - E_{\bar{t}}.$$

The time parametrization

$$\frac{d\bar{s}}{ds} = \frac{1}{r(\bar{t})^2}$$

transforms the last expression into

$$H = \frac{p_{\bar{x}}^2}{2} - \frac{r(t)\mu}{(\bar{x} - (1 - \mu))} - \frac{r(t)(1 - \mu)}{(\bar{x} + \mu)} + \bar{x}\dot{r}(\bar{t})r(\bar{t})p_{\bar{x}} - r(\bar{t})^2 E_{\bar{t}}.$$

Translating

$$\bar{x} \rightarrow \bar{x} + \left(\frac{1}{2} - \mu\right)$$

we get

$$\begin{aligned} H = & \frac{p_{\bar{x}}^2}{2} - \frac{r(t)\mu}{(\bar{x} - \frac{1}{2})} - \frac{r(t)(1 - \mu)}{(\bar{x} + \frac{1}{2})} \\ & + \left(\bar{x} + \frac{1}{2} - \mu\right)\dot{r}(\bar{t})r(\bar{t})p_{\bar{x}} - r(\bar{t})^2 E_{\bar{t}}. \end{aligned} \quad (16)$$

Birkhoff transformation (11) in our one dimensional example ($v = 0$) is given by

$$\begin{aligned} \bar{x} &= \frac{1}{2u} \left(u^2 + \frac{1}{4}\right), \\ \bar{t} &= \tau, \end{aligned} \quad (17)$$

which lifts as

$$\begin{aligned} p_{\bar{x}} &= \frac{8u^2}{4u^2 - 1} p_u, \\ E_{\bar{t}} &= E_{\tau}. \end{aligned}$$

With this transformation (16) becomes after simplification

$$H = \frac{64u^2}{(4u^2 - 1)^2} \frac{p_u^2}{2} - r(\bar{t}) \frac{64u^2 \left[\frac{1}{2u} \left(u^2 + \frac{1}{4}\right) (2\mu - \frac{1}{2}) + \frac{1}{2}\right]}{(4u^2 - 1)^2} + \mathcal{F} - r(\bar{t})^2 E_{\tau},$$

where

$$\mathcal{F} = \dot{r}(\tau)r(\tau) \frac{4u^2 + 1}{4u^2 - 1} u p_u.$$

Doing the reparametrization given by

$$\frac{d\eta}{d\bar{s}} = \frac{(4u^2 - 1)^2}{64u^2}$$

we obtain

$$\begin{aligned} H = & \frac{p_u^2}{2} - r(\tau) \left(\frac{1}{16u} (4u^2 + 1) (4\mu - 1) + \frac{1}{2} \right) \\ & + \dot{r}(\tau)r(\tau) \frac{(16u^4 + 1)}{64u} p_u - r(\tau)^2 E_{\tau}, \end{aligned}$$

which is free of collisional singularities (see Remark 6).

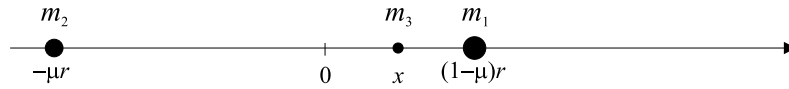


FIGURE 3. Elliptic restricted three-body problem.

Remark 8. The same ideas used in the parabolic case can be used to regularize the binary collisions of the one dimensional Newtonian elliptic restricted three-body problem, where the equal primaries perform elliptic collisions, while the infinitesimal mass body moves in the line between the primaries, during the time between two consecutive elliptic collisions (see Figure 3).

A preliminary study of this problem can be found in [17].

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